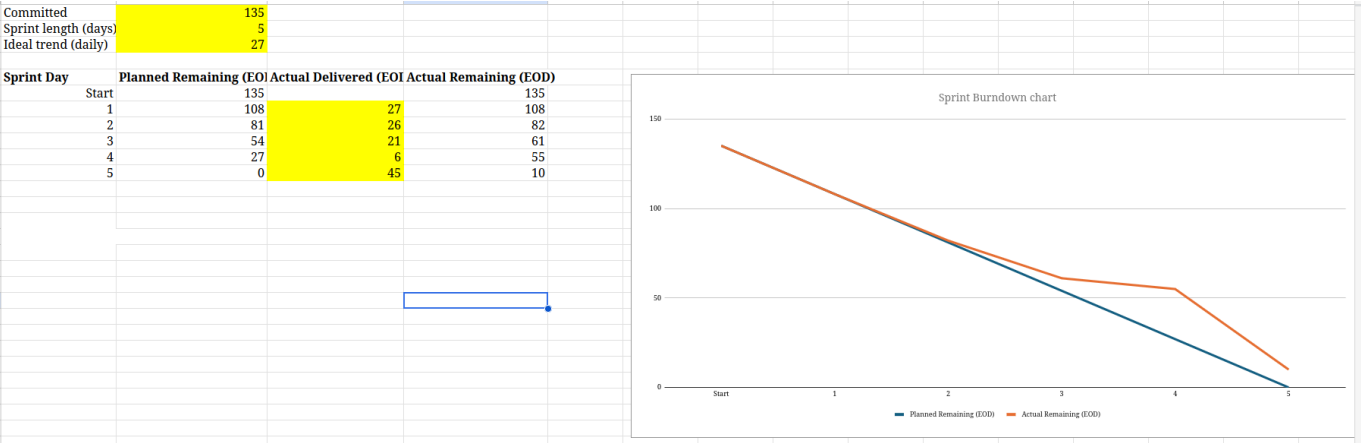


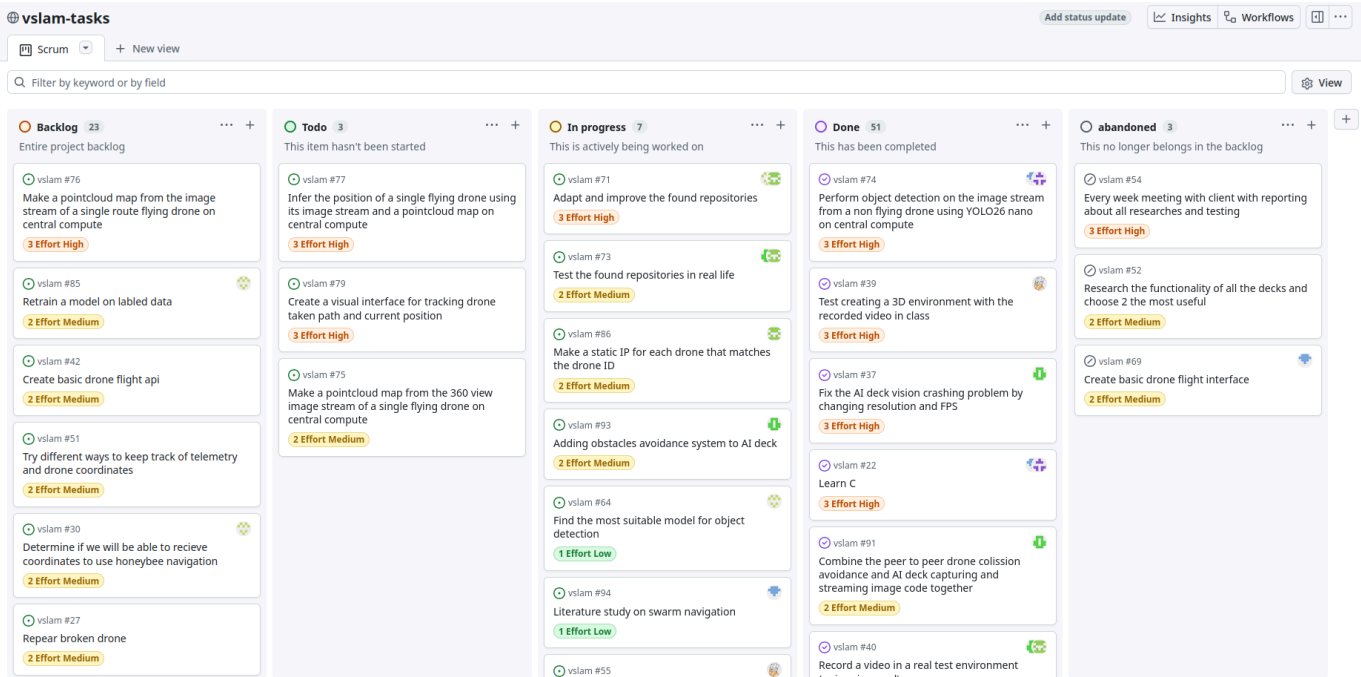
PROJECT MANAGEMENT DOCUMENTATION

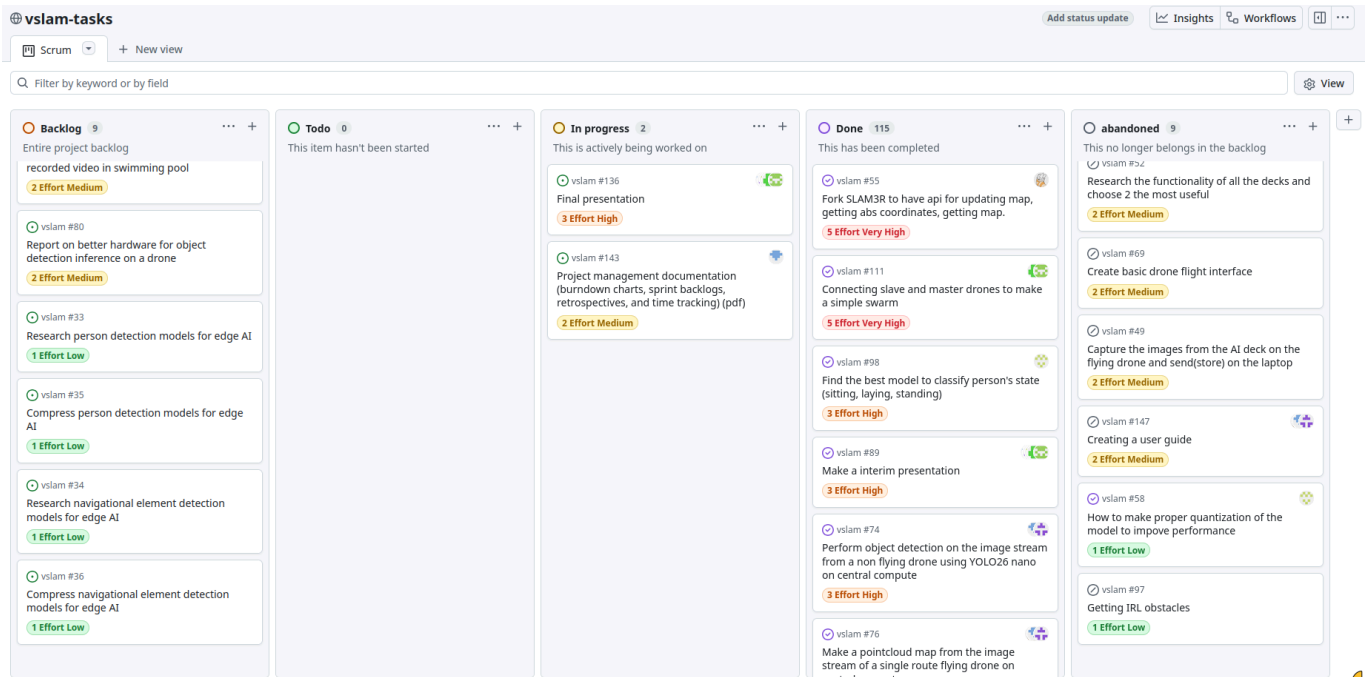
Burndown chart



Sprint backlogs

Interim





Retrospective

Week 01

Summary

During the first week we got a more detailed explanation of the project from our client. We focussed mainly on literature research and testing sample code for all technologies our end result might use.

Positives

During the kickoff presentation all team members contributed to getting a clearer view of our clients demands by asking the right questions. This resulted in a good understanding of our goals and the work needed to accomplish those. Each member had their own area to work on. Everyone was very driven for literature research, checking out existing repositories and initial example code testing. Results followed fast, development went smooth. There was a good team vibe. The progress of week one resulted in image based pointcloud maps, P2P drone communication, image streaming with the AI deck.

Negatives

The quick start did have some drawbacks. Because everyone was very focussed on their individual work, integration and agile principles were neglected. There was no improvement of the project board, backlog and it wasn't updated when working on tasks. This resulted in repetition of tasks. Not all tests and research was added to the github code base. Early good results also prevented looking further into more existing implementations. Systems all work seperate but no systems had been combined yet.

Takeaways

We need more focus on integration, pushing more code to the github repository. We also need a more streamlined way to work together on crazyflie firmware. keep better track of the project board to avoid

repetition of work. Do more research and testing.

Notes

literature research

initial experimentation

Nikita

- + everyone understood goal of the project ,good reading and thinking about implementation
- everyone working too seperated, not applying for combining 2 teams officially

Noah

- + same as nikita, smoother developepment
- repetition of questions, misunderstanding, not properly implementing scrum

Tofa

- + good research on existing solutions, got a map working
- not more thorough looking for implementation

Anton

- + Good progress on P2P communcations, good progress on starting 2 work with ai deck
- not updating code base on github

Nazar

- + good working team vibe
- miscommunication in repetition of work

Roel

- + everyone was very driven in the research and testing phase, a lot of progress was made
- More scrum principles were forgotten about, implementing backlog and doing retrospectives

Week 02

Positives

Fields of research were continued with improvement (P2P, video streaming, object detection, mapping). There was ood communication with third parties to get access to necesarry material and a testing environment. Integration improved making use of forks to write our own firmware.

Negatives

The big bottleneck in week 2 were edge cases with software (flow deck, camera collision avoidance repos) and slow processes like mapping. These edge cases caused a lot of delay. Our project board has

become a bit stale. There are a lot of duplicate or abandoned issues. Our drive for the project has also slightly decreased causing some team members to lose focus during work and others to not work the required 7 hours per day average.

Takeaways

We need a new sense of direction as a group to align our goals better, improve communication and commit more to made agreements. Refreshing our project board with input from all team members

Notes

Noah

- + improved P2P collision avoidance
- wasted a lot of time figuring out ground texture causes confusion in the flow deck

Nikita

- + Combining code and forking the crazyflie firmware for our purposes
- time lost reflashing, rebooting drones, debugging

Tofa

- + mapping conversion to api
- mapping localization, mapping takes a lot of time

Anton

- + better image streaming with personal network (router), environment testing
- problems with figuring out collision avoidance repos, reprogramming of the drones

Nazar

- + getting material and location access
- project board needs improvement, less issues (mainly duplicates, abandoned)

Roel

- + Greater focus on integration
- loss of focus during work hours, people not being present average 7 hours per day

Week 03

Positives

The interim presentation was a great success, everyone was well prepared, everyone had an equal stake and the slides looked nice as well. The addition of the ranger deck to our tools caused a split approach that will allow us to work on more possible solutions at the same time. Mapping systems now allow for the combination of footage for multiple drones.

Negatives

The mapping system does not combine multiple drone pov's well due to each drone image stream scaling differently. There is a drop in motivation and presence from every team member and communication deteriorates there is a lack of daily standups

Takeaways

There needs to be a shift in team dynamics

Notes

Noah

- + smooth presentation, single drone mapping
- communication between the team is lackluster, mainly because of separate working environments, absence of standups

Tofa

- + initial map combination for different drones testable
- different drone footages scale differently and cannot be combined

Nikita

- + start of experiments with the ranger deck drones, good splitting of the teams
- not a lot of motivation because the project is very research heavy

Anton

- + Good presentation good preparation, looking at separate approaches
- team members growing tired of research

Roel

- + Great work on the presentation, looking at separate approaches with ranger deck
- lack of communication between the team members

Week 04

Positives

An official splitting up of teams helps communication between sub group team members, the full team managed to get an overview of our final deliverables. Ranger deck sub groups are working on their approaches. There is progress on the person state detection AI model.

Negatives

The scrum board is not maintained properly. A lot of time is already spent on map combination. The client cannot produce specific deliverables for us to achieve, they are more focussed on research which is not the goal of our project and mentally draining. There is yet again a lack of motivation and presence of team members.

Takeaways

We need to keep the scrum board up to date. Our team will have to come up with the specific deliverables.

Notes

Noah

- + 1 ranger deck 2 ai deck, daily standups
- scrum board maintenance

Tofa

- + good scope for demonstration
- a lot of times spent on working on map combination

Nikita

- + all group memebers have demonstrated results, a lot of approaches were taken and a lot of results were achieved
- the client is very focussed on research and not deliverables wich draining

Anton

- + finished the ai model and discovering problems with detection because of camera quality
- client communication, client absence for meetings

Roel

- + good splitting of teams
- lack of motivation, presence

Week 05

Positives

Most team members are very happy with the end result and also very happy that it's the end of the project this week. We have two demo's that feel complete and can demonstrate our work while also having extensive documentation. Team members can still converse about the project in a friendly and professional matter. We did not take up any new additions from the client and managed to define our own deliverables.

Negatives

Some team members feel the end result to be underwhelming. Others were tired and demotivated this week. There's also team members that are displeased as some team members have not committed fully to this project and showed a lack of dedication.

Notes

Noah

- + Happy with his part of the end result, happy with a complete demo
- Some team memebers not committing fully to the project in this last week

Nikita

- + Very happy to be near the end result and near the end, happy with the amount work
- Less communication between group members

Tofa

- + We finally have an end result, not taking in new changes from the client, maintaining a backlog, we have concrete deliverables, complete documentation
- Feels like we gave up on the project, feels underwhelming

Anton

- + Finishing the project and there is a deliverable result, there was enough time to do all we planned to do
- The least amount of motivation/tiredness

Nazar

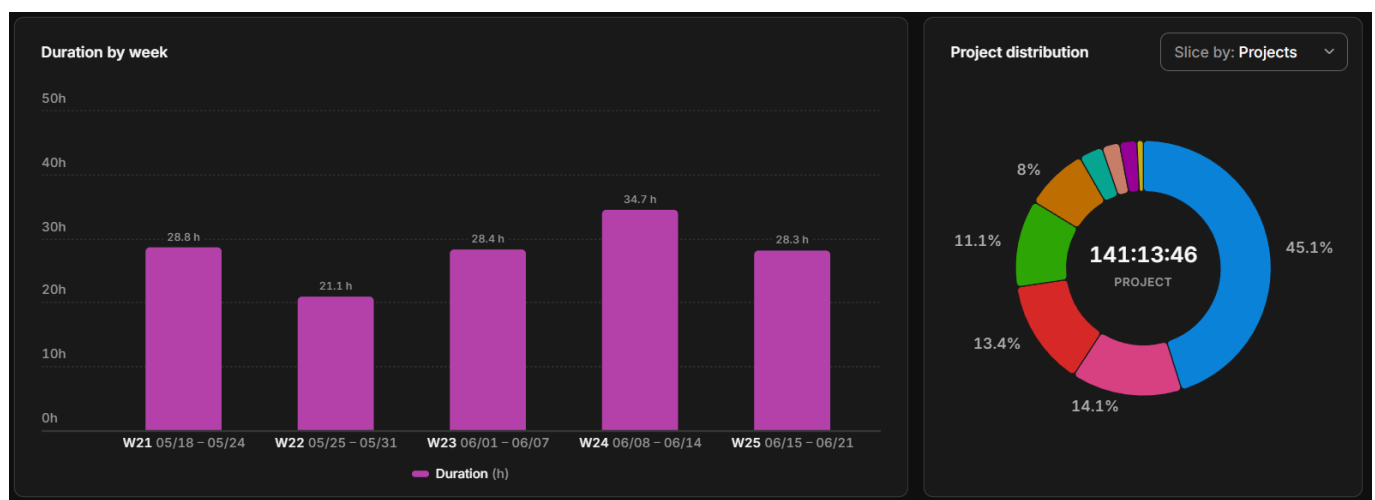
- + Even though we are tired there are still good vibes and everyone can still be friendly
- We became less motivated because it's the finishing weeks

Roel

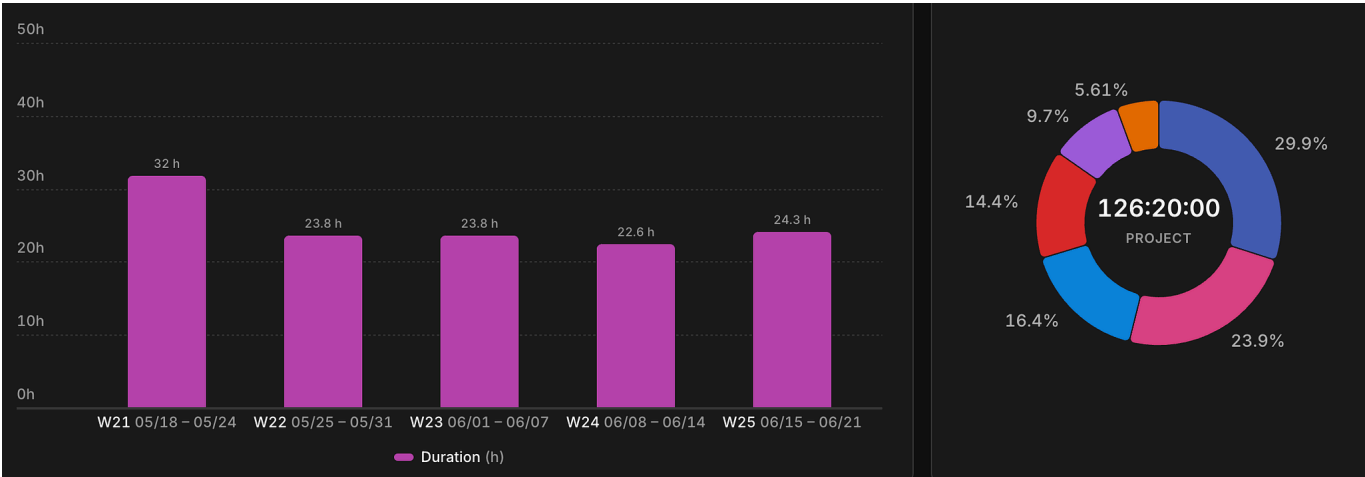
- + All the parts that needed to be integrated together are put together
- Complete lack of professionalism and dedication from some team members which could have improved our end result

Time Tracking

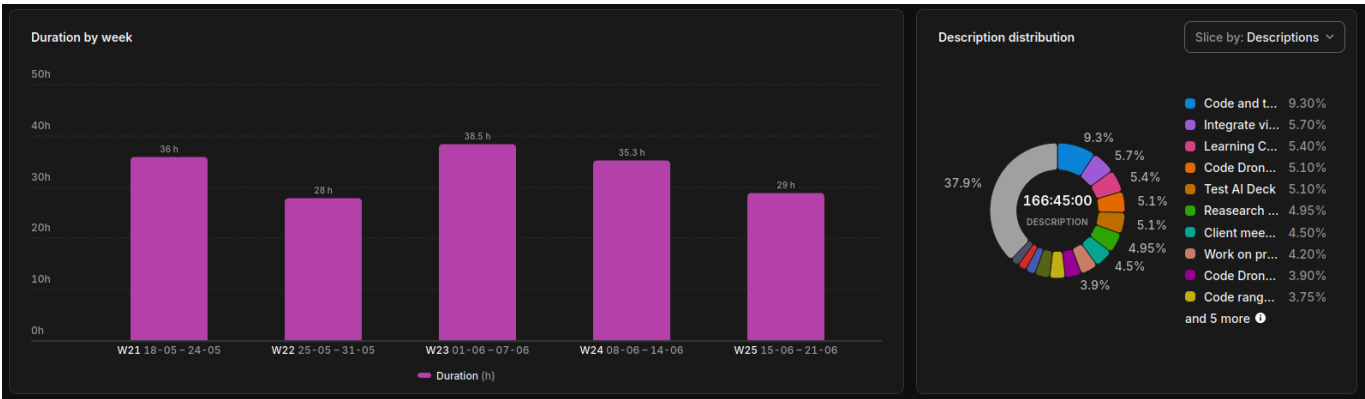
Noah



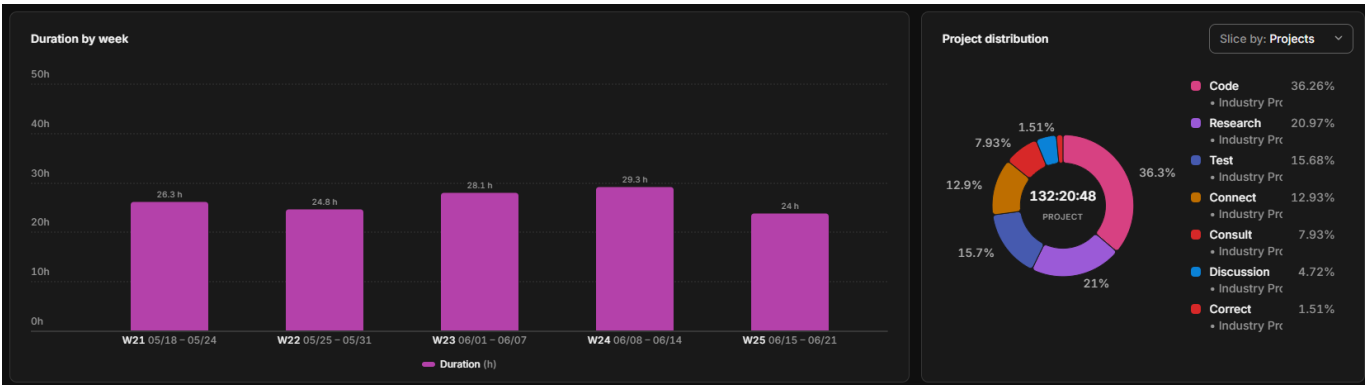
Anton



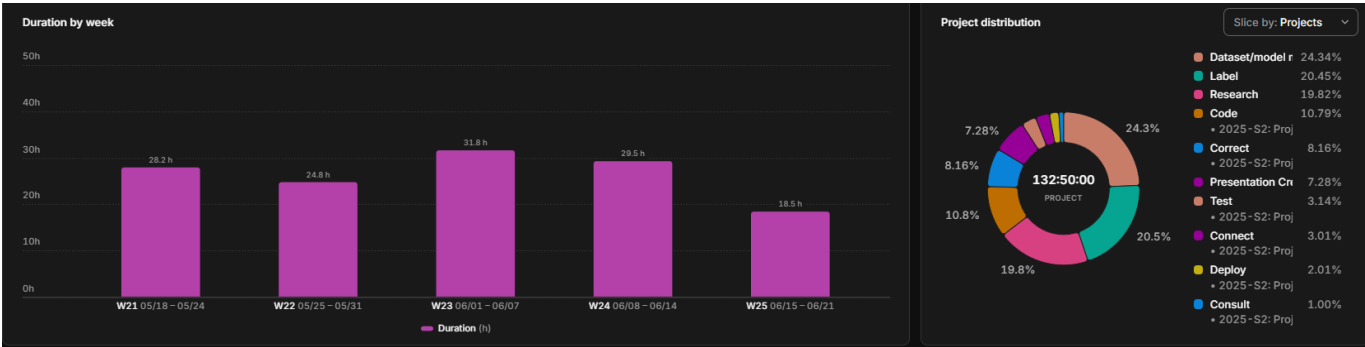
Roel



Nikita



Nazar



Tofa

